

Armin Salmasi

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WORK EXPERIENCE

Software Engineer & Model Developer

2023 - Present

Thermo-Calc Software AB

- Designed and implemented physics-based algorithms and data structures for core calculation engines and simulation workflows.
- Fine-tuned MLIP backends (MACE, Meta's UMA, FAIRChem/OCP) with benchmark datasets (OC20, OC22, Materials Project).
- Engineered agentic AI workflows and multi-agent orchestration systems leveraging LLMs, prompt engineering, and tool-use agents.
- Managed CI/CD pipelines, automated testing, and build environments, ensuring high reliability for production systems.

Key Projects:

- **MLIP Framework Integration:** Fine-tuned and deployed MLIP models within computational pipelines for high-throughput materials screening.
- **GenAI Toolchain:** Engineered agentic GenAI systems and code assistants using RAG, LLMs, and tool-use orchestration.
- **Data Infrastructure:** Developed robust ETL pipelines and deployed scalable database solutions for scientific data.

R&D Simulation Specialist

2023

Epiroc Drilling Tools AB

- Utilized computational modeling and manufacturing data analysis to optimize production parameters and reduce trial-and-error costs.
- Collaborated with cross-functional engineering teams to model phase transformation dynamics under high-pressure conditions.

Research Intern (Master Thesis)

2016

Sandvik Coromant AB

- Applied **Density Functional Theory (DFT)** calculations to screen high-entropy alloy candidates.
- Developed diffusion simulation models to predict gradient formation during sintering.

TECHNICAL SKILLS

Comp. Mat. Science	Thermo-Calc, DICTRA, TC-PRISMA, COMSOL Multiphysics, ICME, CAL-PHAD, ParaView, VTK
Atomistic & MLIPs	LAMMPS, VASP, Molecular Dynamics (MD), DFT, Meta's UMA, MACE, FAIRChem/OCP
Materials Datasets	OC20, OC22, Materials Project, MPDS, Inorganic & Organic Crystal Structures
ML & AI	PyTorch, TensorFlow, MLIP Fine-Tuning, Scikit-learn, LLMs (RAG, Prompt Eng.), LangChain, MLFlow
Languages	Python (Advanced), Java, Fortran, CUDA, SQL, Bash Script
Data & Analytics	Data Analysis, Descriptive & Inferential Statistics, Hypothesis Testing, SciPy (Stats), Pandas, NumPy, MongoDB, MySQL, Seaborn, Matplotlib, Plotly
HPC & Distributed	MPI (Fortran), OpenMP (Fortran), mpi4py, SLURM, Dask, PySpark, Celery, Multiprocessing, Threading, Asyncio, Concurrent
Platform & DevOps	GCP, Docker, Kubernetes, GitLab CI/CD, GitHub Actions, Jenkins, Git, ETL
OS	Linux, macOS, Windows
IDEs & Copilots	IntelliJ, PyCharm, VSCode, Cursor, Antigravity, Claude Code, Codex

CERTIFICATIONS & SELECTED COURSEWORK

- **Applied AI & Data Science Program**, MIT (Ongoing, Batch 2026)
- **Supervised ML: Regression & Classification**, DeepLearning.AI
- **Improving Deep Neural Networks**, DeepLearning.AI
- **Neural Networks & Deep Learning**, DeepLearning.AI
- **Structuring Machine Learning Projects**, DeepLearning.AI
- **Practical Deep Learning (HPC)**, ENCCS
- **CAMD Summer School (DFT)**, DTU Physics
- **CSC & PDC HPC Summer Schools**, Solvalla/KTH
- **RACIRI Summer School on Synchrotron X-Ray & Neutron Diffraction**, Rügen
- **CodeRefinery (Software Development)**, NeIC
- **Software Tools: ML to Phase Diagrams**, MGF
- **Advanced Computational Techniques & Modeling**

EDUCATION

Ph.D., Engineering Materials Science

KTH Royal Institute of Technology

Focus: ICME, Mass Transport Simulation, and Numerical Modeling. Served as a Teaching Assistant for multiple advanced master's level and PhD courses.

M.Sc., Materials Science

KTH Royal Institute of Technology

Focus: Simulation of Gradient Formation using Thermodynamic and Kinetic modeling.

M.Sc., Nano Materials

Materials & Energy Research Center

Focus: Thin Film Deposition and Characterization.

B.S., Extractive Metallurgy

Amirkabir University of Technology

SELECTED PUBLICATIONS

 [Google Scholar Profile](#) —  [DIVA Portal Profile](#)

1. **Holmström, E., Lizárraga, R., Linder, D., Salmasi, A., et al.** (2018). High entropy alloys: Substituting for cobalt in cutting edge technology. *Applied Materials Today*, 12, 322–329.
2. **Lindvall, R., Bjerke, A., Salmasi, A., et al.** (2023). Predicting wear mechanisms of ultra-hard tooling in machining Ti6Al4V by diffusion couples and simulation. *J. Eur. Ceram. Soc.*, 43(2), 291–303.
3. **Graves, A., Salmasi, A., Graham, S. J., et al.** (2023). Chemical interactions during machining and in diffusion couples. *Wear*, 516–517, 204604.
4. **Salmasi, A., Graham, S. J., Galbraith, I., et al.** (2021). Mobilities of Ti and Fe in disordered TiFe-BCC. *CALPHAD*, 74, 102300.
5. **Pouranvari, M., Ghasemi, A., & Salmasi, A.** (2022). Solidification time during TLP bonding of Ni-based superalloys. *Metall. Mater. Trans. A*, 53(1), 126–135.
6. **Salmasi, A., Blomqvist, A., & Larsson, H.** (2019). Microstructural evolution and mechanical properties of graded carbides. *Results in Materials*, 1, 100008.
7. **Salmasi, A., Hanif, I., Jonsson, T., & Larsson, H.** (2023). Liquid phase migration in cemented carbides. *Int. J. Refract. Met. Hard Mater.*, 112, 106118.
8. **Salmasi, A., Forslund, A., & Ruban, A. V.** (2022). Viscosity of liquid cobalt from ab-initio molecular dynamics (*manuscript*).